BHARTI SCHOOL OF ENGINEERING AND COMPUTER SCIENCE

Design LU based Forensic Network for Reconstructing VoIP (Voice over IP) Calls using Wireshark.

CPSC_5207EL_07 Computer Network Forensics

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Chapter 1 – Project Synopsis, and Literature Review.

1.1. Project Synopsis:

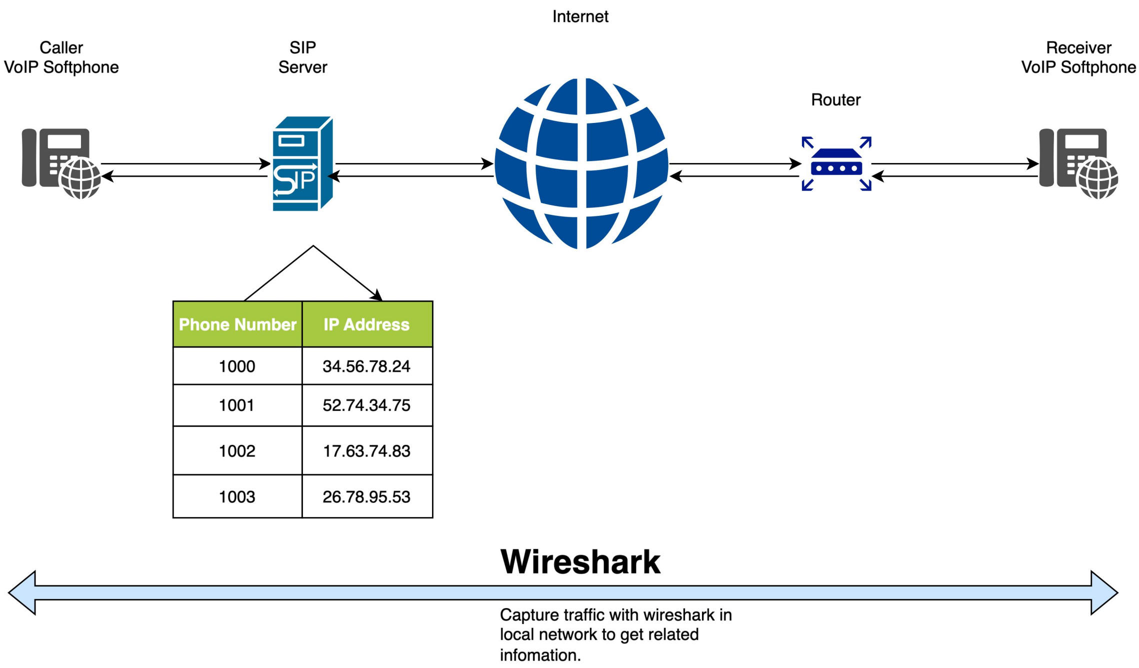
- **Project Abstract and Literature Review**:

VoIP is widely used in remote conferencing, intelligent customer service, cross-border Internet phone calls, and other fields due to its low cost, cross-platform, strong scalability, and the ability to combine AI to achieve smarter analysis and decision-making. However, it also brings the risk of being attacked by hackers, such as session hijacking, caller ID spoofing, and vishing. The project involves designing a forensic network for reconstructing VoIP calls using Wireshark, a sophisticated network protocol analyzer. By capturing and analyzing VoIP packets, we can recover call-related information, identify potential issues impacting call quality, and reconstruct audio from voice data if available. Additionally, we will provide an overview of the security threats associated with VoIP traffic and propose best practices for encrypting VoIP traffic to mitigate these risks.

- **Projects vision and goals/objectives:**
    - **Capture Traffic**: Use wireshark to capture the VoIP traffic.
    - **Filter Data**: Apply SIP and RTP filters to effectively isolate pertinent call data streams.
    - **Extract Caller and Receiver Information**: Analyze SIP packets in detail to extract comprehensive caller and receiver information.
    - **Analyze Call Quality**: Utilize packet timestamps to identify potential issues impacting call quality and provide insights into network performance.
    - **Reconstruct Audio**: If applicable, attempt to reconstruct audio from RTP packets based on available voice data.
    - **Summarized Security Threats**: Evaluate the security risks associated with VoIP traffic and propose encryption measures to mitigate these threats.

- **System Flowchart:**

1) Overview Flow



Phone Number	IP Address
1000	34.56.78.24
1001	52.74.34.75
1002	17.63.74.83
1003	26.78.95.53

Fig. 1. Overview VoIP workflow.

2) VoIP Protocol Stack

We used UDP/SIP/SDP/RTP for VoIP Protocol. And we focus on the main protocol here SIP and RTP.

Layer 7	Application	RTP/RTCP (Real-time Transport Protocol)	User interface and application logic
Layer 6	Presentation	SDP (Session Description Protocol)	Data format conversion and encryption
Layer 5	Session	SIP (Session Initiation Protocol)	Establish, manage, and terminate sessions
Layer 4	Transport	UDP (User Datagram Protocol)	End-to-end reliable or efficient transmission
Layer 3	Network	IP (Internet Protocol)	Addressing and routing
Layer 2	Data Link	Ethernet, WiFi, etc.	Physical addressing and frame transmission
Layer 1	Physical	Network cables, wireless signals, etc.	Physical medium and signal transmission

Fig. 2. VoIP Protocol Stack.

3) SIP Dialogue Flow

SIP is a signaling protocol used for initiating, maintaining, and terminating communication sessions.

The general dialogue between the Caller and Receiver is as below.

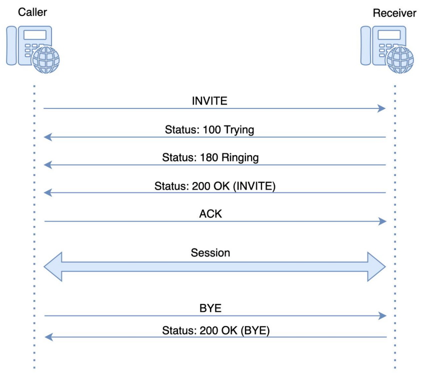


Fig. 3. SIP dialogue flow.

4) RTP Dialogue Flow

RTP is a network protocol for delivering audio and video over IP networks. It transfers the voice to digital signals put them into different chunks(packages), and then delivers them to the receiver.

Dialogue flow as below.

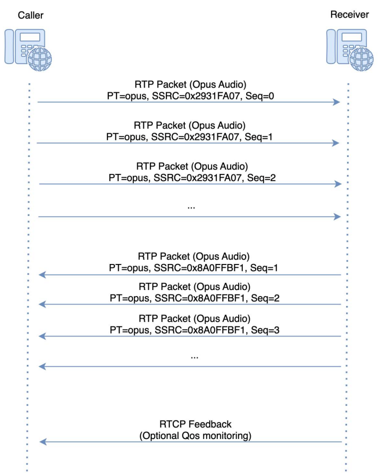


Fig. 4. RTP dialogue flow.

Chapter 2 – System Model.

2.1. Technology Stack and System Requirements:

- **Project requirements:**

1) Function Requirements

- Capture traffic: Capture traffic with Wireshark.
- Filter traffic: Apply SIP, RTP, or VoIP filters to isolate relevant call data streams.
- Extract caller and receiver: Extract information about the caller and receiver.
- Analyze call quality: Analyze packet timestamps to detect potential issues affecting call quality.
- Reconstruct audio: Attempt audio reconstruction if voice data is available within the packets.
- Summarize risk: Summarize security risks associated with VoIP traffic and recommend encryption practices.

2) Non-Function Requirements

- More VoIP protocal: SIP and RTP protocal is used in this project, other VoIP protocals have not been included.
- Large scale: Demo for the forensic analysis process, large scale traffic has not been included.

3) Hardware Requirements

- Minimum required: RAM>=4GB, recommended: >=8GB for better performance.
- Health network: Stable network is needed to set up VoIP calls.
- Storage: Need storage space to save related files.

- **Technologies/Tools Stack**:
  - **Wireshark:** Software to capture and analyze internet traffic.
  - **Linphone:** Software to simulate VoIP softphone calls on a SIP server.
  - **SIP and RTP filter:** Capture VoIP traffic in Wireshark.
  - **Audio Reconstruction Tools:** Decode and replay the audio in Wireshark.

- **Project Scheduling and Tasks/Function Distribution:**
  - **Sprint 1(June 03 -June 05):** Project background research, distribute group roles, project division of roles, ice-breaking for team members.
  - **Sprint 2(June 05 -June 12):** Design for the implementation, set up document outline, breakdown tasks, set up environment, capture traffic, call flow diagrams, design PPT, architecture design.
  - **Sprint 3(June 12 -June 19)**: Create workflow steps, filter applied, extract caller and callee information, supply SIP/RTP analysis notes, describe tools
  - **Sprint 4(June 19 -June 25)**: Quality analysis, reconstructed audio, create VoIP protocol stack, complete implementation steps. Draw enhanced screenshots.
  - **Sprint 5(June 26 -June 29)**: Review all delivered material and distribute the presentation slides.
  - **Sprint 6(June 30 -July 10)**: Collaborative presentation process.

2.2. DFDs, Class Diagrams and Entity Relationships (ERs):

- **Project Data Flow Diagrams (DFDs):**

The DFD illustrates the primary data flow for VoIP from the call setup to the result generated.

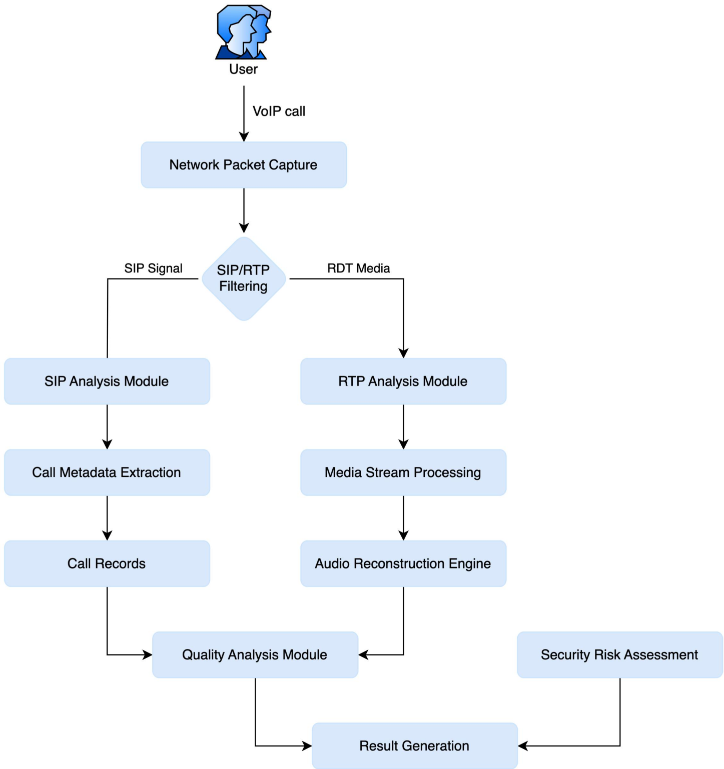


Fig. 5. Project data flow diagrams.

- **Project Class Diagrams:**

The Class diagram illustrates the system's object-oriented architecture. It is currently implemented by Wireshark.

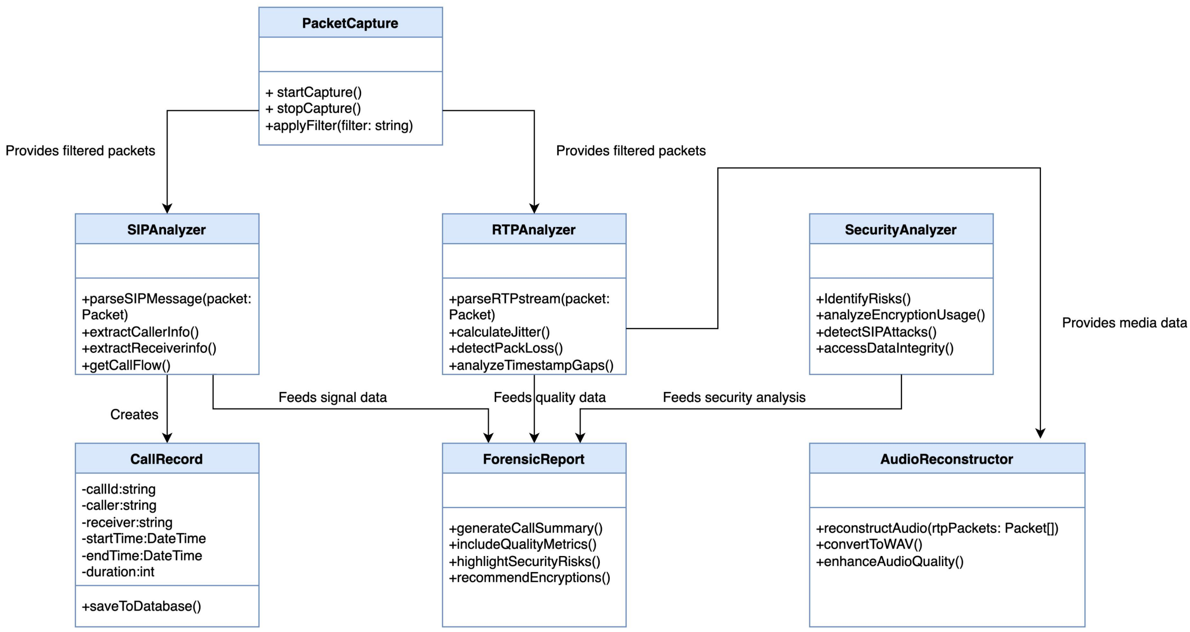


Fig. 6. Project class diagrams.

- **Project ERs:**

The ERs below show the main properties of the key action and reflect the relationship between them.

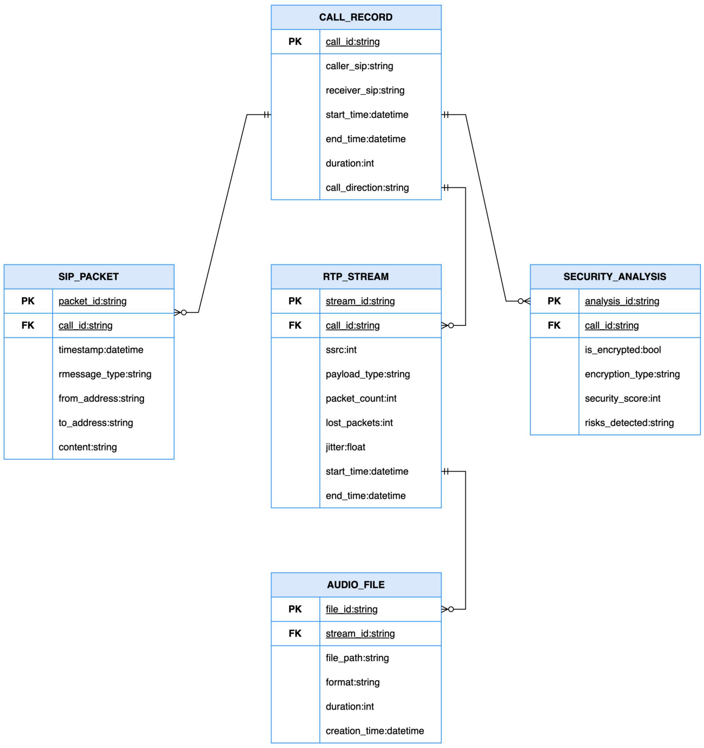


Fig. 7. Project ERs.

Chapter 3 – Implementation, Results and Conclusion.

3.1. System Methods/Functions Implementation:

- **All Methods/Functions Implementation:**

Overview steps for implementing steps.

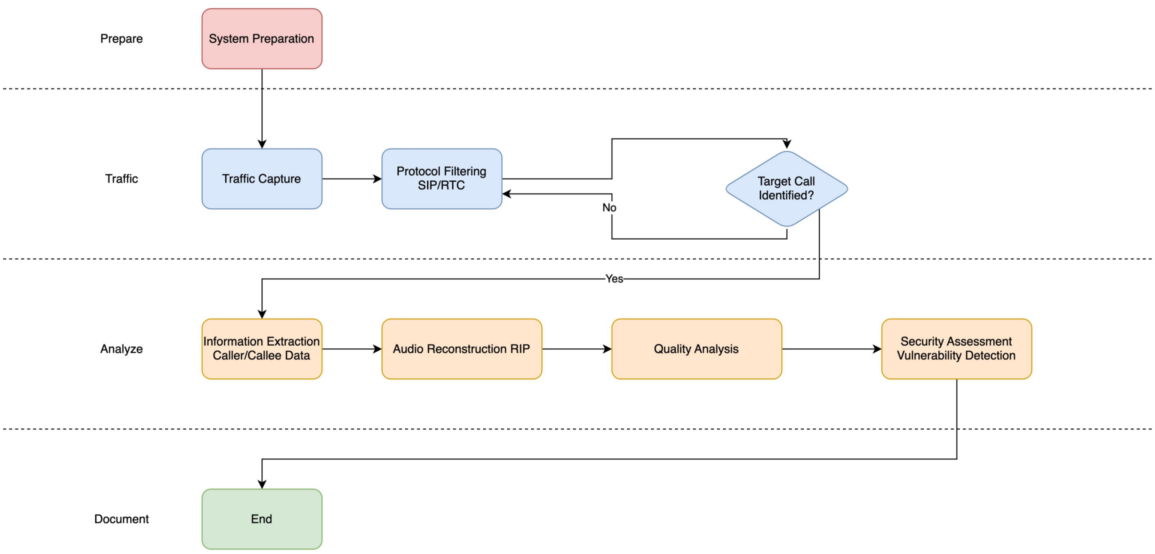


Fig. 8. Implement steps.

1) Setup the Environment

- **SIP server register:** Register VoiP.ms and set up two accounts for two phone numbers.
- **Linkphone Installation and Configuration:** Install Linkphone on sender and receiver equipment.
- **Network Configuration**: Ensure two equipment are on the same network(Wi-Fi etc.) for packet capture.

2) Filter Traffic to get SIP and RTP packages

- **Capture traffic**: Capture traffic with Wireshark.
- **SIP Filter:** Apply the SIP filter in Wireshark to get the SIP package.Input 'SIP' in the filter and get the sip-related packages.
- **RTP Filter:** Apply the RTP filter in Wireshark to get the package.Input 'rtp' in the filter and get the rtp-related packages.

3) Extract Caller and Receiver

- Click on the **invite** record.
- Check the details panel to get the caller and receiver information from the SIP invite package.

4) Analyze Call Quality

- Call Flow
    - Click the "Telephony > VoIP Calls" menu.
    - Click "Flow sequence" to show SIP flow.

Fig. 9. Snapshot of call flow.

- Analyze RTP Signaling
    - Enable RTP protocol: Analyze -> Enabled Protocols -> select RTP

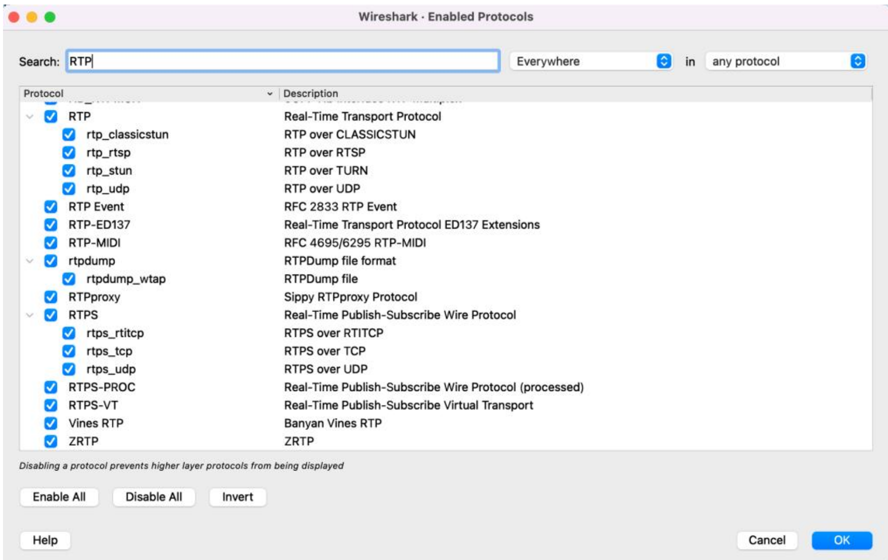


Fig. 10. Snapshot to enable RTP protocols.

- Go to “Telephony > RTP > RTP Streams.” it will display all RTP records.
- Analyze the key metrics
    - Packet Loss:Missing packets = 0% <1%
    - Jitter: Packet delay variation, Max Jitter=39.53ms > 30ms
    - Latency: Delay in transmission, Max Delta > 150ms one-way

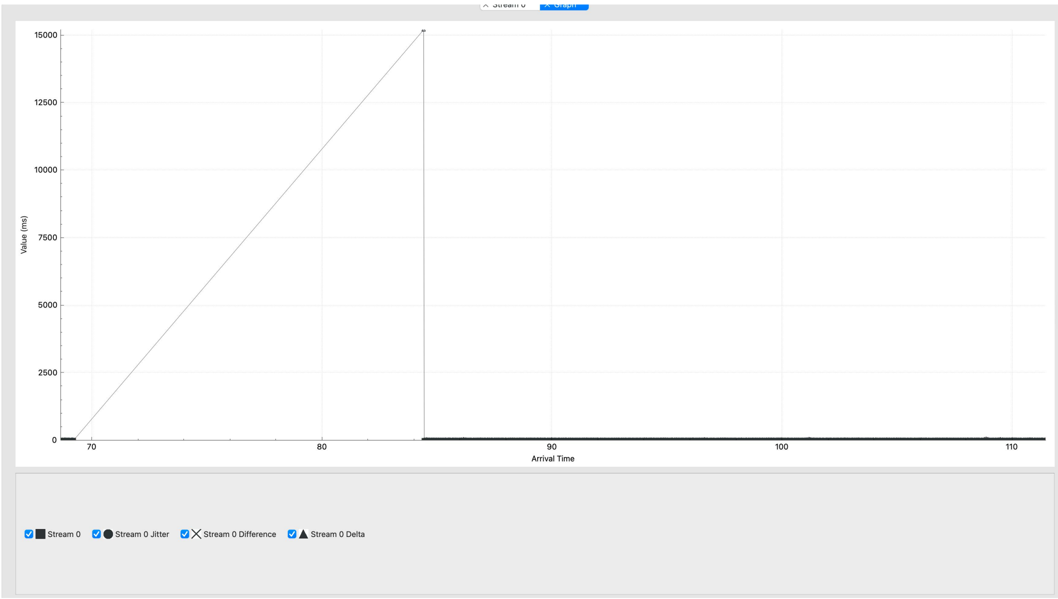


Fig. 11. Snapshot of RTP caller to receiver voice quality.

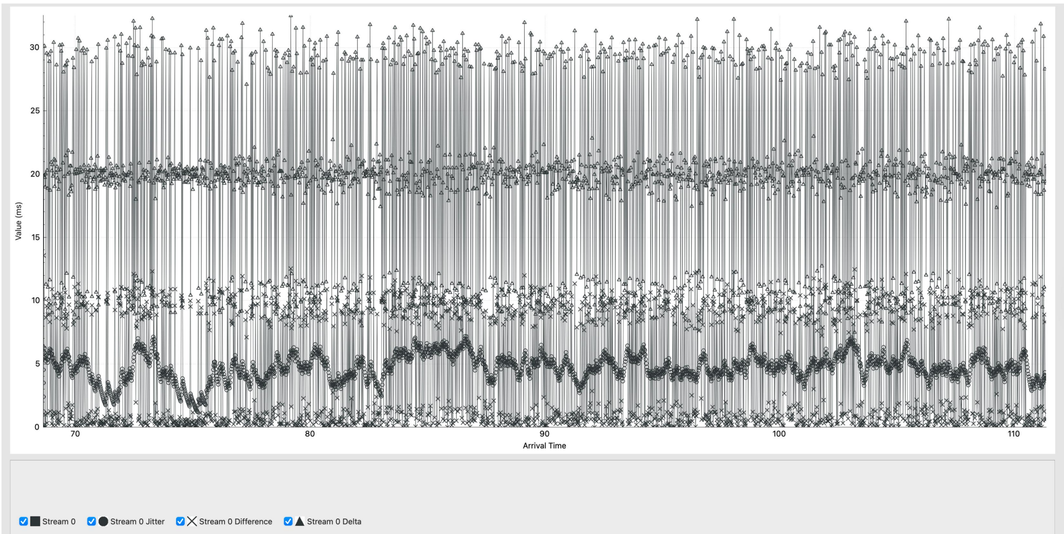


Fig. 12. Snapshot of RTP receiver to caller voice quality.

5) Reconstruction Audio

- Click on the **invite** record.
- Go to "Telephony".
- Select "VoIP Calls" and then click the "Play Streams" button. And then it will replay the voice. If the RTP packages have not been decoded, we can decode the UDP packages and then replay them.

6) Security Analysis

- Eavesdropping Detection - Not Good.
  - Encrypted identification by payload type check.
    - Click “Telephony > RTP > RTP Streams.”
    - Check the payload type and codec info to identify if it has been encrypted.
- Tamper Detection
  - SIP Attack Identification - OK.
    - SIP abnormal statuses are identified by clicking the “Telephony > SIP Statistics” menu. Check if there is an unexcepted status.
  - RTP Identification - OK.
    - Click “Telephony > RTP > RTP Streams.”
    - Check sequence number is continuous.

3.2. System Results:

- **Project results/Outputs (all methods/functions):**

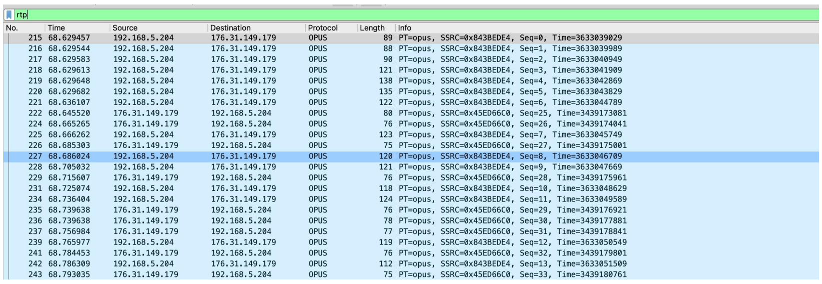
1) SIP filter result

sip

No.	Time	Source	Destination	Protocol	Length	Info
87	59.994204	192.168.5.204	176.31.149.179	SIP/SDP	1033	Request: INVITE sip:hafizabdullah@sip.linphone.org |
88	60.157275	176.31.149.179	192.168.5.204	SIP	444	Status: 100 Trying |
93	61.590825	176.31.149.179	192.168.5.204	SIP	517	Status: 180 Ringing |
213	68.147331	176.31.149.179	192.168.5.204	SIP/SDP	1052	Status: 200 Ok (INVITE) |
214	68.627223	192.168.5.204	176.31.149.179	SIP	874	Request: ACK sip:hafizabdullah@sip.linphone.org;gr=urn:uuid:b22c52f4-c327-4b9b-8275-24351acafff0 |
1314	84.116866	192.168.5.204	176.31.149.179	SIP/SDP	893	Request: INVITE sip:hafizabdullah@sip.linphone.org;gr=urn:uuid:b22c52f4-c327-4b9b-8275-24351acafff0, in-dialog |
1338	84.468646	176.31.149.179	192.168.5.204	SIP/SDP	205	Status: 200 Ok (INVITE) |
1342	84.507629	192.168.5.204	176.31.149.179	SIP	923	Request: ACK sip:hafizabdullah@sip.linphone.org;gr=urn:uuid:b22c52f4-c327-4b9b-8275-24351acafff0 |
4170	111.382985	192.168.5.204	176.31.149.179	SIP	923	Request: BYE sip:hafizabdullah@sip.linphone.org;gr=urn:uuid:b22c52f4-c327-4b9b-8275-24351acafff0 |
4177	111.611728	176.31.149.179	192.168.5.204	SIP	461	Status: 200 Ok (BYE) |

Fig. 13. Snapshot of SIP filter.

2) RTP filter result



rtp

No.	Time	Source	Destination	Protocol	Length	Info
215	68.629457	192.168.5.204	176.31.149.179	OPUS	89	PT=opus, SSRC=0x843BEDE4, Seq=0, Time=3633039029
216	68.629544	192.168.5.204	176.31.149.179	OPUS	88	PT=opus, SSRC=0x843BEDE4, Seq=1, Time=3633039989
217	68.629583	192.168.5.204	176.31.149.179	OPUS	90	PT=opus, SSRC=0x843BEDE4, Seq=2, Time=3633040949
218	68.629613	192.168.5.204	176.31.149.179	OPUS	121	PT=opus, SSRC=0x843BEDE4, Seq=3, Time=3633041909
219	68.629648	192.168.5.204	176.31.149.179	OPUS	138	PT=opus, SSRC=0x843BEDE4, Seq=4, Time=3633042869
220	68.629682	192.168.5.204	176.31.149.179	OPUS	135	PT=opus, SSRC=0x843BEDE4, Seq=5, Time=3633043829
221	68.636107	192.168.5.204	176.31.149.179	OPUS	122	PT=opus, SSRC=0x843BEDE4, Seq=6, Time=3633044789
222	68.645520	176.31.149.179	192.168.5.204	OPUS	80	PT=opus, SSRC=0x45ED66C0, Seq=25, Time=3439173081
224	68.665265	176.31.149.179	192.168.5.204	OPUS	76	PT=opus, SSRC=0x45ED66C0, Seq=26, Time=3439174041
225	68.666262	192.168.5.204	176.31.149.179	OPUS	123	PT=opus, SSRC=0x843BEDE4, Seq=7, Time=3633045749
226	68.685303	176.31.149.179	192.168.5.204	OPUS	75	PT=opus, SSRC=0x45ED66C0, Seq=27, Time=3439175001
227	68.686024	192.168.5.204	176.31.149.179	OPUS	120	PT=opus, SSRC=0x843BEDE4, Seq=8, Time=3633046709
228	68.705032	192.168.5.204	176.31.149.179	OPUS	121	PT=opus, SSRC=0x843BEDE4, Seq=9, Time=3633047669
229	68.715607	176.31.149.179	192.168.5.204	OPUS	76	PT=opus, SSRC=0x45ED66C0, Seq=28, Time=3439175961
231	68.725074	192.168.5.204	176.31.149.179	OPUS	118	PT=opus, SSRC=0x843BEDE4, Seq=10, Time=3633048629
234	68.736404	192.168.5.204	176.31.149.179	OPUS	124	PT=opus, SSRC=0x843BEDE4, Seq=11, Time=3633049589
235	68.739638	176.31.149.179	192.168.5.204	OPUS	76	PT=opus, SSRC=0x45ED66C0, Seq=29, Time=3439176921
236	68.739638	176.31.149.179	192.168.5.204	OPUS	78	PT=opus, SSRC=0x45ED66C0, Seq=30, Time=3439177881
237	68.756984	176.31.149.179	192.168.5.204	OPUS	77	PT=opus, SSRC=0x45ED66C0, Seq=31, Time=3439178841
239	68.765977	192.168.5.204	176.31.149.179	OPUS	119	PT=opus, SSRC=0x843BEDE4, Seq=12, Time=3633050549
241	68.784453	176.31.149.179	192.168.5.204	OPUS	76	PT=opus, SSRC=0x45ED66C0, Seq=32, Time=3439179801
242	68.786309	192.168.5.204	176.31.149.179	OPUS	112	PT=opus, SSRC=0x843BEDE4, Seq=13, Time=3633051509
243	68.793035	176.31.149.179	192.168.5.204	OPUS	75	PT=opus, SSRC=0x45ED66C0, Seq=33, Time=3439180761

Fig. 14. Snapshot of RTP filter.

3) Caller and Receiver Information

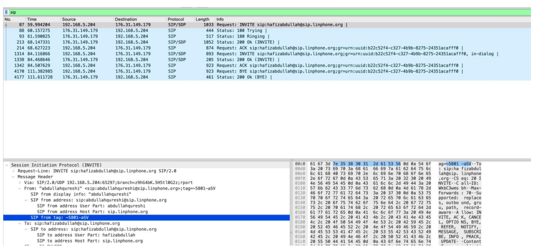


Fig. 15. Snapshot of caller and receiver information.

4) Quality Analyze Result

Destination Address	Destination Port	SSRC	Start Time	Duration	Payload	Packets	Lost	Min Delta (ms)	Mean Delta (ms)	Max Delta (ms)	Min Jitter	Mean Jitter	Max Jitter
192.168.5.204	65526	0x45ed66c0	68.645520	42.82	opus	1383	759 (35.4%)	0.000000	30.983295	15198.512000	0.015938	6.713032	10.726937
176.31.149.179	13382	0x843bede4	68.629457	42.74	opus	2143	0 (0.0%)	0.030000	19.951468	32.539000	1.211372	4.719263	7.195726

Fig. 16. Snapshot of quality analyze information.

5) Reconstruction Audio

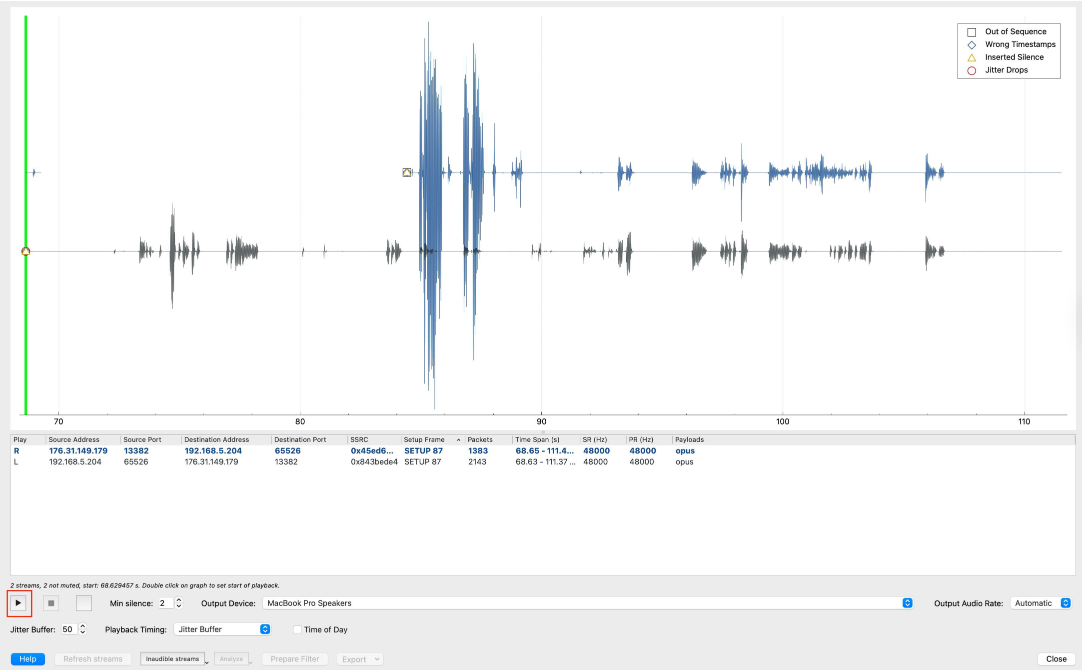


Fig. 17. Snapshot of play streams.

6) Security Analysis Result

- SIP Attack Identification - OK.

Request Method / Response Code	Count	Resent	Min Setup (s)	Avg Setup (s)	Max Setup (s)
SIP Requests					
ACK	2	0	0.390000	4.512000	8.634000
BYE	1	0	0.000000	0.000000	0.000000
CANCEL	0	0	0.000000	0.000000	0.000000
DO	0	0	0.000000	0.000000	0.000000
INFO	0	0	0.000000	0.000000	0.000000
INVITE	2	0	0.000000	0.000000	0.000000
MESSAGE	0	0	0.000000	0.000000	0.000000
NOTIFY	0	0	0.000000	0.000000	0.000000
OPTIONS	0	0	0.000000	0.000000	0.000000
PRACK	0	0	0.000000	0.000000	0.000000
PUBLISH	0	0	0.000000	0.000000	0.000000
QAUTH	0	0	0.000000	0.000000	0.000000
REFER	0	0	0.000000	0.000000	0.000000
REGISTER	0	0	0.000000	0.000000	0.000000
SPRACK	0	0	0.000000	0.000000	0.000000
SUBSCRIBE	0	0	0.000000	0.000000	0.000000
UPDATE	0	0	0.000000	0.000000	0.000000
SIP Responses					
100 Trying	1	0	0.000000	0.000000	0.000000
180 Ringing	1	0	0.000000	0.000000	0.000000
181 Call Is Being Forwarded	0	0	0.000000	0.000000	0.000000
182 Queued	0	0	0.000000	0.000000	0.000000
183 Session Progress	0	0	0.000000	0.000000	0.000000
199 Informational - Others	0	0	0.000000	0.000000	0.000000
200 OK	3	0	0.000000	0.000000	0.000000
202 Accepted	0	0	0.000000	0.000000	0.000000
204 No Notification	0	0	0.000000	0.000000	0.000000
299 Success - Others	0	0	0.000000	0.000000	0.000000
300 Multiple Choices	0	0	0.000000	0.000000	0.000000
301 Moved Permanently	0	0	0.000000	0.000000	0.000000
302 Moved Temporarily	0	0	0.000000	0.000000	0.000000
305 Use Proxy	0	0	0.000000	0.000000	0.000000
380 Alternative Service	0	0	0.000000	0.000000	0.000000
399 Redirection - Others	0	0	0.000000	0.000000	0.000000
400 Bad Request	0	0	0.000000	0.000000	0.000000
401 Unauthorized	0	0	0.000000	0.000000	0.000000
402 Payment Required	0	0	0.000000	0.000000	0.000000
403 Forbidden	0	0	0.000000	0.000000	0.000000
404 Not Found	0	0	0.000000	0.000000	0.000000
405 Method Not Allowed	0	0	0.000000	0.000000	0.000000
406 Not Acceptable	0	0	0.000000	0.000000	0.000000
407 Proxy Authentication Required	0	0	0.000000	0.000000	0.000000
408 Request Timeout	0	0	0.000000	0.000000	0.000000
410 Gone	0	0	0.000000	0.000000	0.000000
412 Conditional Request Failed	0	0	0.000000	0.000000	0.000000
413 Request Entity Too Large	0	0	0.000000	0.000000	0.000000

Fig. 18. Snapshot of stastic of SIP status.

- **System Results/Output Constraints/Parameters:**
  i). Operational/Transitional Efficiency:

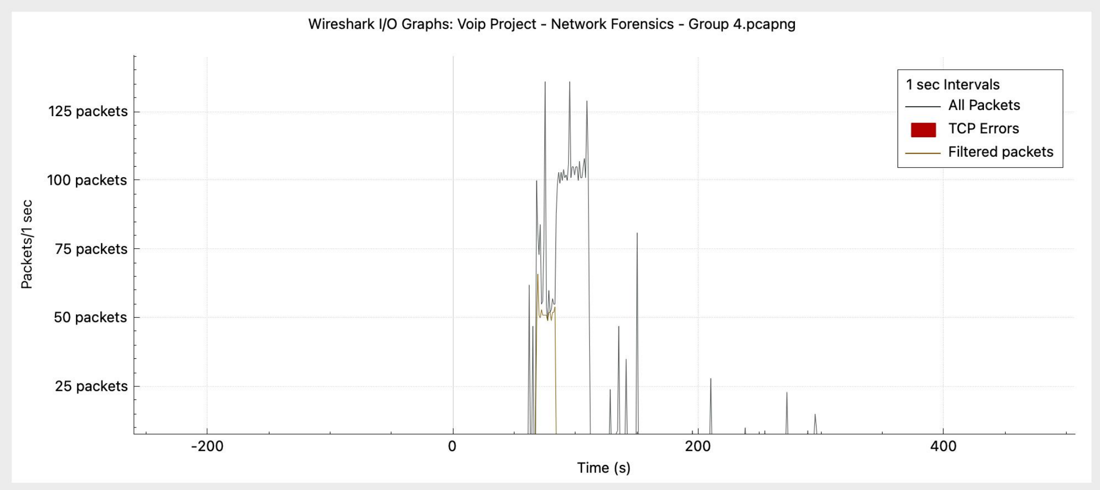


Fig. 19. Snapshot of stastic of I/O graphs.

ii). Risk Assessment:

- Unencrypted calls have the following risks
  - Wiretapping: Get VoIP call with unencrypted protocol may lead to the secretly monitor.
- Recommendations
  - Update the encryped protocols, for example, use SRTP to encrypt RTP media streams and use TLS to secure SIP signaling channels.

iii). System Reliability and Downtime: No packets lost.
iv). Data Security & Privacy: Unencrypted packets lead to potential risk.

3.3. Conclusion:

- **Conclusion:**

This project effectively demonstrates the reconstruction of VoIP calls using Wireshark and underscores the security vulnerabilities associated with unencrypted VoIP communications. By capturing and analyzing SIP and RTP data packets, we successfully extracted call metadata, identified issues affecting call quality, and reconstructed audio content from the RTP stream. The research results highlight the urgent need for encryption mechanisms and secure VoIP protocols to reduce risks such as eavesdropping, call hijacking and data tampering.

- **Future Enhancements:**
  - **Process efficiency improvement**: The entire process covers most of the common basic steps of VoIP network forensics, and can be highly automated through Python code + AI Agent. After the optimization is completed, various situations in the actual forensics process can be continuously iterated to make the processing process more intelligent to complete unsupervised automatic forensics.
  - **Security improvement**: This process does not encrypt the call, and there are multiple risks such as call tampering and information leakage. The call can be encrypted later to ensure user data security.

- **Functional improvement**: This process does not simulate the forensics steps in the attack scenario. Later, the forensics process and results in the attack scenario can be simulated based on historical attack cases.
- **Performance improvement**: This process mainly uses wireshare's own functions, but when large-scale data packets appear, the built-in functions cannot support the analysis of large amounts of data. Python programming can be used to assist in analysis.

- **References:**

- Document
  - Wireshark User Guide: https://www.wireshark.org/docs/wsug_html_chunked/
  - Wireshark for VoIP calls: https://wiki.wireshark.org/VoIP_calls
  - SIP and RTP Analyze Guide: https://support.yeastar.com/hc/en-us/articles/360007606533-How-to-Analyze-SIP-Calls-in-Wireshark
  - VoIP Analysis Fundamentals with Wireshark: https://sharkfest.wireshark.org/retrospective/sfus/presentations12/BI-7_VoIP_Analysis_Fundamentals.pdf
  - Linphone Documentation: https://www.linphone.org/en/docs-category/bien_demarrer-en/
- Youtube Tutorial:
  - Voip Process: https://www.youtube.com/watch?v=n_b2sJP2pBk
  - Voip wireshark Process: https://www.youtube.com/watch?v=uZI9ZnKRudg